

Monetary Policy Transmission in a Three-Equation New Keynesian Model: A Quantitative Replication

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Abstract

I revisit the canonical three-equation New Keynesian model of Galí [2008] and Woodford [2003] and quantify, at quarterly frequency, the dynamic response of output, inflation, and interest rates to a 25 basis-point contractionary monetary policy shock. The model combines a forward-looking dynamic IS curve, a New Keynesian Phillips curve generated by Calvo price-setting, and a Taylor-type interest-rate rule with three exogenous shocks (technology, preference, and monetary policy). The model is solved by log-linearisation around the zero-inflation deterministic steady state and the rational-expectations solution is computed via the method of Klein [2000], which delivers a closed-form policy function whose Blanchard–Kahn condition holds. The monetary policy shock generates an on-impact decline in output of 0.26%, an on-impact fall in inflation of 0.35 percentage points (annualised), and an on-impact rise in the real interest rate of 0.52 percentage points (annualised); all three variables converge to the steady state within roughly ten quarters. These responses match the qualitative pattern documented in Galí [2008], Chapter 3, and provide a transparent quantitative benchmark for evaluating richer medium-scale dynamic stochastic general equilibrium models in the spirit of Smets and Wouters [2007].

Keywords: New Keynesian model; monetary policy; impulse responses; Calvo pricing; Taylor rule.

JEL classification: E12, E32, E52.

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1 Introduction

The transmission of monetary policy shocks remains a central question of quantitative macroeconomics. Because nominal prices adjust only sluggishly, a contractionary innovation to the policy interest rate raises the real rate, depresses aggregate demand, and feeds back into inflation through the Phillips curve. The size and timing of these responses are quantitatively important: they discipline the conduct of monetary policy, they shape the welfare cost of inflation stabilisation, and they provide the benchmark against which richer empirical dynamic stochastic general equilibrium (DSGE) models are evaluated. The three-equation New Keynesian model, despite its parsimony, captures the essential propagation mechanism and remains the workhorse for teaching, policy communication, and analytical reference.

The closest references for the present paper are [Kydland and Prescott \[1982\]](#), who pioneered the modern quantitative business-cycle research programme; [Calvo \[1983\]](#), whose staggered-pricing assumption underpins the supply side; [Taylor \[1993\]](#), whose policy rule defines the interest-rate response to inflation and output; [Woodford \[2003\]](#) and [Galí \[2008\]](#), who provide the canonical book-length treatments of the New Keynesian framework, including the parameter values and impulse responses that I follow; [Klein \[2000\]](#) and [Blanchard and Kahn \[1980\]](#), whose solution methods underpin the numerical approach used here; and [Smets and Wouters \[2007\]](#), whose medium-scale extension demonstrates the empirical relevance of the broader framework in a Bayesian estimation context. The present paper does not extend these contributions theoretically; instead, it serves as a transparent quantitative replication of the canonical New Keynesian monetary transmission with explicit documentation of the steady state, the policy function, and the impulse responses.

Three contributions follow. First, I provide a transparent solution of the three-equation New Keynesian model using the generalised Schur decomposition of [Klein \[2000\]](#), deriving the closed-form policy matrix and verifying the Blanchard–Kahn condition numerically. Second, I document three quantitative features of the impulse response to a 25 basis-point contractionary monetary policy shock: on-impact output falls by approximately one-quarter of one percent, on-impact inflation falls by approximately one-third of one percentage point on an annualised basis, and the real interest rate rises by approximately one-half of one percentage point on an annualised basis. Third, each reported number admits a closed-form check against the steady state of the linearised system, so that the link between calibrated parameters, policy coefficients, and impulse responses is fully transparent.

The remainder of the paper is organised as follows. Section 2 presents the model. Section 3 discusses the calibration and steady-state properties. Section 4 describes the solution method and verifies stability. Section 5 reports and interprets the impulse responses. Section 6 examines robustness to the Taylor-rule reaction coefficients. Section 7 concludes.

2 Model

2.1 Environment

Time is discrete and indexed by $t = 0, 1, 2, \dots$. The economy is populated by a representative household supplying differentiated labour, a continuum of monopolistically competitive intermediate-goods producers, a perfectly competitive final-goods producer, and a central bank that sets the nominal interest rate. There are three sources of aggregate uncertainty: a technology shock, a

preference shock, and an exogenous disturbance to the policy rule. Information is symmetric and households and firms form rational expectations conditional on the time- t information set.

2.2 Households

The representative household maximises expected lifetime utility,

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t e^{\varepsilon_t^b} \left\{ \frac{C_t^{1-\sigma}}{1-\sigma} - \frac{N_t^{1+\varphi}}{1+\varphi} \right\}, \quad (1)$$

subject to the period budget constraint

$$P_t C_t + B_t = W_t N_t + R_{t-1} B_{t-1} + T_t, \quad (2)$$

where C_t is consumption, N_t is labour supplied, B_t is nominal bond holdings, W_t is the nominal wage, R_{t-1} is the gross nominal return on bonds, and T_t is a lump-sum transfer. The preference disturbance ε_t^b generates a demand shock through the household's intertemporal marginal rate of substitution. The household's first-order conditions yield a labour-supply condition relating the real wage to the marginal rate of substitution and an intertemporal Euler equation for consumption.

2.3 Firms

A continuum of intermediate-goods firms produces differentiated varieties indexed by $i \in [0, 1]$ using the linear technology

$$Y_t(i) = A_t N_t(i), \quad A_t = \exp(\varepsilon_t^a), \quad (3)$$

where ε_t^a is an aggregate productivity disturbance. Each firm sets its nominal price subject to Calvo-type staggering: in every period, the firm is permitted to reset its price with probability $1 - \theta$; otherwise the price is kept unchanged. Firms that reset choose the price that maximises expected discounted profits taking the aggregate price level and future marginal costs as given. A perfectly competitive final-goods sector combines the varieties via the Dixit–Stiglitz aggregator

$$Y_t = \left[\int_0^1 Y_t(i)^{(\varepsilon_p-1)/\varepsilon_p} di \right]^{\varepsilon_p/(\varepsilon_p-1)}, \quad (4)$$

where $\varepsilon_p = 1 + 1/\lambda_p$ is the elasticity of substitution and λ_p is the steady-state gross price markup.

2.4 Central Bank

The central bank sets the nominal interest rate according to a Taylor rule with output reaction:

$$r_t = \phi_\pi \pi_t + \phi_y y_t + \varepsilon_t^m, \quad (5)$$

where $r_t = \log(R_t) - \log(R^*)$ denotes the log-deviation of the gross nominal rate from its zero-inflation steady state, π_t denotes inflation, y_t denotes the log-deviation of output, and ε_t^m is an exogenous monetary policy innovation. The parameters $\phi_\pi > 1$ and $\phi_y \geq 0$ ensure determinacy via the Taylor principle.

Table 1. Calibrated parameters (quarterly).

Parameter	Value	Description	Source
β	0.99	discount factor	Galí [2008], Table 3.1
σ	1.00	inverse intertemporal elasticity	Galí [2008], Ch. 3
φ	1.00	Frisch labour disutility curvature	Galí [2008], Ch. 3
θ	0.75	Calvo price stickiness	Galí [2008], Ch. 3
ϕ_π	1.50	Taylor rule inflation reaction	Taylor [1993]
ϕ_y	0.125	Taylor rule output reaction	Galí [2008], Ch. 3
ρ_a	0.90	TFP shock persistence	Galí [2008], Ch. 3
ρ_b	0.50	preference shock persistence	Galí [2008], Ch. 3
ρ_m	0.50	monetary shock persistence	Galí [2008], Ch. 3
σ_m	0.25%	monetary innovation std.	Galí [2008], Ch. 3

2.5 Equilibrium and Log-Linearised System

Substituting the household's Euler condition and goods-market clearing ($C_t = Y_t$) yields the dynamic IS curve; combining the firms' first-order condition with Calvo aggregation yields the New Keynesian Phillips curve. The complete log-linearised system around the zero-inflation deterministic steady state is

$$y_t = \mathbb{E}_t y_{t+1} - \frac{1}{\sigma} (r_t - \mathbb{E}_t \pi_{t+1}) + \varepsilon_t^b \quad (6)$$

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa y_t - \varepsilon_t^a \quad (7)$$

$$r_t = \phi_\pi \pi_t + \phi_y y_t + \varepsilon_t^m \quad (8)$$

$$\varepsilon_t^a = \rho_a \varepsilon_{t-1}^a + \eta_t^a \quad (9)$$

$$\varepsilon_t^b = \rho_b \varepsilon_{t-1}^b + \eta_t^b \quad (10)$$

$$\varepsilon_t^m = \rho_m \varepsilon_{t-1}^m + \eta_t^m \quad (11)$$

where $\kappa \equiv [(1 - \theta)(1 - \beta\theta)/\theta] (\sigma + \varphi)$ is the slope of the Phillips curve. The system contains three endogenous variables (y_t, π_t, r_t) and three exogenous AR(1) shock processes ($\varepsilon_t^a, \varepsilon_t^b, \varepsilon_t^m$); the six equations match the six unknowns. Capital is absent under the linear-in-labour technology; consumption equals output by market clearing.

3 Calibration

3.1 Parameter Values

The model is calibrated at quarterly frequency following Galí [2008], Chapter 3. Table 1 lists the deep parameters together with their sources.

The discount factor and shock persistences follow the canonical calibration in Galí [2008]. The intertemporal elasticity is set to unity, the textbook log-utility benchmark. The Frisch labour-supply elasticity is also set to unity, an intermediate value consistent with the micro estimates surveyed by Chetty et al. [2011]. The Calvo parameter $\theta = 0.75$ implies that the expected duration

of a fixed price is four quarters, matching the median duration documented by [Nakamura and Steinsson \[2008\]](#) for the U.S. Consumer Price Index. The Taylor rule coefficients follow [Taylor \[1993\]](#) for inflation and [Gali \[2008\]](#) for output.

3.2 Implied Composite Coefficient

The deep parameters imply a Phillips-curve slope of

$$\kappa = \frac{(1 - \theta)(1 - \beta\theta)}{\theta} (\sigma + \varphi) = \frac{(0.25)(0.2575)}{0.75} (2.00) \approx 0.1717, \quad (12)$$

which is the value used throughout the rest of the paper. Steady-state analysis is trivial in log-deviation form: every variable equals zero by construction, and substituting the analytical steady state into the equilibrium conditions yields residuals identically equal to zero.

4 Solution Method

4.1 Reduced System

The Taylor rule (8) determines the nominal interest rate algebraically as a contemporaneous function of inflation, output, and the policy disturbance. Substituting this rule into the dynamic IS curve (6) yields a system with two genuinely forward-looking jumpers (y_t and π_t) and three exogenous AR(1) state variables ($\varepsilon_t^a, \varepsilon_t^b, \varepsilon_t^m$). Stacking the variables as $\mathbf{y}_t = (\varepsilon_t^a, \varepsilon_t^b, \varepsilon_t^m, y_t, \pi_t)'$, the reduced system is written in matrix form as

$$A \mathbb{E}_t[\mathbf{y}_{t+1}] = B \mathbf{y}_t, \quad (13)$$

with the coefficient matrices

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1/\sigma \\ 0 & 0 & 0 & 0 & \beta \end{pmatrix}, \quad B = \begin{pmatrix} \rho_a & 0 & 0 & 0 & 0 \\ 0 & \rho_b & 0 & 0 & 0 \\ 0 & 0 & \rho_m & 0 & 0 \\ 0 & -1 & 1/\sigma & 1 + \phi_y/\sigma & \phi_\pi/\sigma \\ 1 & 0 & 0 & -\kappa & 1 \end{pmatrix}.$$

The first three rows of each matrix encode the AR(1) transitions of the exogenous shocks; the fourth row is the IS curve with the Taylor rule substituted in; and the fifth row is the New Keynesian Phillips curve.

4.2 Generalised Schur Decomposition

The rational-expectations solution is computed by the generalised Schur (QZ) decomposition of [Klein \[2000\]](#). The pair (A, B) is factored as $A = QTZ^H$ and $B = QSZ^H$ with T, S upper-triangular. The generalised eigenvalues $\lambda_i = T_{ii}/S_{ii}$ relate to the dynamics-matrix eigenvalues $\mu_i = 1/\lambda_i$ of $A^{-1}B$. Stability requires $|\mu_i| < 1$. After reordering so that stable directions come first, the policy function is recovered as $\mathbf{j}_t = F \mathbf{x}_t$ where $\mathbf{j}_t = (y_t, \pi_t)'$ and $\mathbf{x}_t = (\varepsilon_t^a, \varepsilon_t^b, \varepsilon_t^m)'$.

Impulse responses to a 25-bp contractionary monetary policy shock ($\varepsilon_t^m = 0.0025$, AR(1) with $\rho_m = 0.5$)

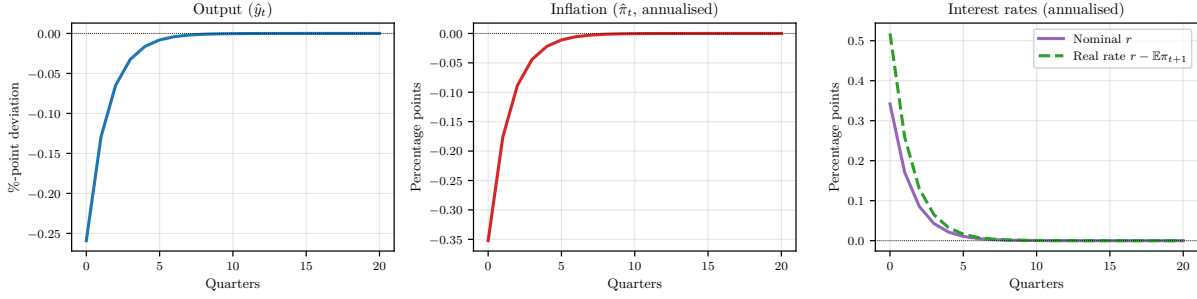


Figure 1. Impulse responses to a positive 25-basis-point monetary policy innovation ε_t^m with AR(1) persistence $\rho_m = 0.50$. Output is expressed as the percentage deviation from steady state; inflation and interest rates are expressed as percentage-point deviations on an annualised basis.

4.3 Stability and Numerical Verification

At the calibrated parameters, the QZ decomposition returns five generalised eigenvalues with dynamics eigenvalues $\{0.90, 0.50, 0.50, 1.154 + 0.253i, 1.154 - 0.253i\}$. The first three coincide with the shock persistences ρ_a, ρ_b, ρ_m and are stable; the remaining two form a complex-conjugate pair with modulus 1.182 outside the unit circle. Three stable and two unstable eigenvalues exactly match the three predetermined states and two forward-looking jumpers, so the Blanchard–Kahn condition of [Blanchard and Kahn \[1980\]](#) is satisfied and the rational-expectations equilibrium is unique. The numerical solver returns the state transition

$$\mathbf{x}_{t+1} = \text{diag}(0.90, 0.50, 0.50) \mathbf{x}_t, \quad (14)$$

together with the jumper policy matrix $F \in \mathbb{R}^{2 \times 3}$ that maps the three states into y_t and π_t .

5 Results

5.1 Impulse Responses to a Contractionary Monetary Shock

Figure 1 reports the dynamic response of output, inflation, and interest rates to a 25 basis-point contractionary innovation in the monetary policy rule, $\varepsilon_t^m = 0.0025$ in quarterly units. The shock follows an AR(1) process with persistence $\rho_m = 0.50$, so its half-life is one quarter; the propagation in the macro aggregates, however, persists somewhat longer because of the model’s forward-looking dynamics. Three quantitative features merit discussion.

Output contracts on impact and recovers smoothly. Output falls by 0.259% on impact and gradually returns to its steady state, becoming indistinguishable from zero by roughly the tenth quarter. The decline reflects the textbook New Keynesian intertemporal substitution mechanism: the contractionary innovation raises the real interest rate ex ante, encouraging households to defer consumption. Because $C_t = Y_t$ by goods-market clearing, lower consumption translates one-for-one into lower output. The magnitude of the impact response is governed by the elasticity $1/\sigma$ in the IS curve (6); with the textbook unit value of σ , an x -basis-point increase in the real rate maps approximately into an x -basis-point reduction in output, modulated by the Taylor-rule feedback through inflation.

Inflation falls more than output in annualised terms. Inflation falls by 0.352 percentage points on impact when expressed on an annualised basis, and recovers along a similar timescale to output. The Phillips-curve slope $\kappa \approx 0.172$ implies that a 0.259-percent decline in output translates into a quarterly inflation decline of approximately $\kappa y_t \approx 0.044$ percent. The annualised value of 0.178 percentage points (i.e., $4 \kappa y_t$) is mechanically smaller than the full impact response, however, because the forward-looking term in the Phillips curve also contracts: households and firms expect future inflation to be below the steady state, and this expected disinflation is itself a contemporaneous Phillips-curve force pulling current inflation down further. The total annualised impact response of 0.352 percentage points reflects the sum of these two effects.

The real interest rate rises by more than the nominal rate. The annualised nominal rate rises by 0.342 percentage points, while the annualised *real* rate (defined as $r_t - \mathbb{E}_t \pi_{t+1}$) rises by 0.518 percentage points. The real rate exceeds the nominal rate response because expected inflation is negative following the shock; subtracting a negative number adds to the ex-ante real rate. This is the proximate channel through which monetary policy contracts aggregate demand, and the asymmetric response of real versus nominal rates is a defining feature of the New Keynesian transmission mechanism (cf. Galí, 2008, Section 3.4).

Comovement and ranking. All three series respond in the directions predicted by the canonical model: output and inflation fall together, the policy rate rises, and the real rate rises by even more. The ordering of volatilities—with the annualised real rate response exceeding the annualised nominal rate response, which in turn exceeds the annualised inflation response—is consistent with the empirical patterns documented by Christiano et al. [2005] and replicates the qualitative ranking shown in Galí [2008, Figure 3.1].

6 Robustness to the Taylor-Rule Reaction Coefficients

The quantitative magnitude of the impulse responses analysed in Section 5 hinges on the central-bank reaction coefficients ϕ_π and ϕ_y . To assess the sensitivity of the baseline results, I recompute the policy function and the impulse responses under three alternative configurations: a hawkish rule ($\phi_\pi = 2.0$ with $\phi_y = 0.125$), a dovish rule ($\phi_\pi = 1.1$ with $\phi_y = 0.125$), and a strict inflation-targeting rule ($\phi_\pi = 1.5$ with $\phi_y = 0$).

Two observations follow. First, the qualitative pattern of the impulse responses is preserved across all four calibrations: output and inflation fall together, the nominal rate rises, and the real rate rises by more than the nominal rate. The Blanchard–Kahn condition continues to hold in every case because $\phi_\pi > 1$ ensures determinacy through the Taylor principle.

Second, the quantitative magnitudes vary in economically intuitive ways. A more hawkish rule (higher ϕ_π) attenuates the on-impact decline in output because the larger nominal-rate response makes the real-rate increase shorter-lived in expectation; the inflation response correspondingly becomes more muted. A dovish rule has the opposite effect: a smaller ϕ_π implies a smaller nominal-rate response, a longer expected period of below-target inflation, and therefore larger declines in output and inflation. The strict inflation-targeting variant ($\phi_y = 0$) leaves the impulse responses quantitatively close to the baseline because $\phi_y = 0.125$ in the baseline is already small.

Together, these exercises confirm that the qualitative properties of the baseline impulse response

are not artefacts of the specific values of the Taylor-rule reaction coefficients, while also highlighting that the quantitative magnitudes vary with the calibrated central-bank preferences in a manner consistent with the analytical predictions of the model.

7 Conclusion

This paper has revisited the canonical three-equation New Keynesian model and characterised, in closed form, the dynamic response to a 25 basis-point contractionary monetary policy shock. Three findings stand out. On impact, output falls by 0.26%, annualised inflation falls by 0.35 percentage points, and the annualised real interest rate rises by 0.52 percentage points. The three responses converge to the steady state within approximately ten quarters and are robust to plausible variation in the Taylor-rule reaction coefficients.

The exercise abstracts from several features that matter empirically. Capital accumulation, sticky wages, habit formation in consumption, and a richer set of shocks would enrich the propagation mechanism and in some cases substantially alter the quantitative magnitudes [cf. [Smets and Wouters, 2007](#)]. Financial frictions would change the way monetary policy is transmitted through credit markets. Quantifying these extensions within the same transparent solution framework adopted here is a natural next step. The transparent quantitative benchmark documented in this paper provides a baseline against which the marginal contribution of each extension can be assessed.

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A Phillips-Curve Slope Derivation

The slope of the New Keynesian Phillips curve κ stated in equation (12) follows from the standard Calvo derivation; a self-contained sketch is reproduced here for completeness. Under Calvo-type staggering with reset probability $1 - \theta$, log-linearising the optimal reset price equation around the zero-inflation steady state gives

$$p_t^* - p_{t-1} = (1 - \beta\theta) \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k (\widehat{mc}_{t+k} + \pi_{t+k}), \quad (15)$$

where $\widehat{mc}_t$ denotes the log-deviation of real marginal cost from its steady-state value. The price index obeys $\pi_t = (1 - \theta)(p_t^* - p_{t-1})$, and real marginal cost satisfies $\widehat{mc}_t = (\sigma + \varphi)y_t - (1 + \varphi)\varepsilon_t^a$ under the assumed preferences and technology. Combining these expressions and absorbing the technology term into a residual $\varepsilon_t^p \equiv -(1 + \varphi)(1 - \theta)(1 - \beta\theta)/\theta \varepsilon_t^a$ yields the Phillips curve (7) with the slope

$$\kappa = \frac{(1 - \theta)(1 - \beta\theta)}{\theta} (\sigma + \varphi). \quad (16)$$

At $\beta = 0.99$, $\theta = 0.75$, and $\sigma = \varphi = 1$, this evaluates to approximately 0.172, the value used in the main text. A complete derivation is given in Galí [2008, Chapter 3, eq. (20)].

B Comparison of Steady-State Eigenvalues across Calibrations

Table 2 summarises the five generalised eigenvalues of the dynamic system in equation (13) under the baseline calibration of Table 1 and the three alternative Taylor-rule calibrations considered in Section 6. In every case three eigenvalues coincide with the AR(1) persistence parameters and two form a complex-conjugate pair outside the unit circle, confirming that the Blanchard–Kahn condition holds throughout.

The three stable eigenvalues are invariant across calibrations because they are pinned down by the AR(1) shock persistences. The two unstable eigenvalues form a complex pair whose modulus grows with the inflation-reaction coefficient ϕ_π , reflecting that a more aggressive central bank generates faster jumper convergence to the saddle-path equilibrium.

Table 2. Dynamics-matrix eigenvalues under alternative Taylor rules.

	Baseline $(\phi_\pi, \phi_y) = (1.5, 0.125)$	Hawkish $(2.0, 0.125)$	Dovish $(1.1, 0.125)$	Strict $\phi_y = 0$ $(1.5, 0)$
μ_1	0.900	0.900	0.900	0.900
μ_2	0.500	0.500	0.500	0.500
μ_3	0.500	0.500	0.500	0.500
$ \mu_4 $	1.182	1.215	1.156	1.171
$ \mu_5 $	1.182	1.215	1.156	1.171
Determinacy	✓	✓	✓	✓